

Clock Generation and Program Counter (for 8-bit Computer)

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Formerly DA-IICT

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1. Objective

To design and implement the clock circuit and Program Counter (PC) for the 8-bit SAP-1 computer, including:

- A 555 timer-based clock operating in *monostable (single pulse)*, *astable (continuous pulse)*, and *bistable (toggle)* modes.
 - A 4-bit synchronous Program Counter using the 74LS161 IC to track instruction addresses.
 - Integration of both modules with the RAM, ALU, and registers for coordinated system operation.
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2. Materials Required

2.1 For Clock Circuit

- ICs
 - NE555 Timer IC $\times 3$
 - 74LS04 – Hex Inverter ($\times 1$)
 - 74LS32 – Quad 2-input OR Gate IC ($\times 1$)
 - 74LS08 – Quad 2-input AND Gate IC ($\times 1$)
- Resistors
 - $1\text{k}\Omega$ ($\times 2$)
 - $100\text{k}\Omega$ ($\times 2$)
- Capacitors
 - $0.01\mu\text{F}$ ($\times 1$)
 - $0.1\mu\text{F}$ ($\times 1$)
 - $1\mu\text{F}$ ($\times 1$)
- Other Components
 - Potentiometer ($10\text{k}\Omega$)
 - LEDs ($\times 2$)
 - Push button switch
 - Toggle switch
 - Breadboard
 - Connecting wires and cutters
 - Power supply
 - Digital Storage Oscilloscope (DSO)
 - Multimeter

2.2 For Program Counter

- ICs
 - 74LS161 – 4-bit Synchronous Binary Counter
 - 74LS245 – Octal Bus Transceiver
- Resistors
 - 220Ω (for current limiting, $\times 4$)
- Capacitors
 - $0.1\mu\text{F}$ (Decoupling capacitors)
- Other Components
 - LEDs ($\times 4$)
 - Breadboard
 - Power supply

3. Theory

3.1 Clock Circuit

The 555 timer IC was used as the core for generating the clock signal in three different modes:

(a) Astable Mode:

- No external trigger is required.
- The output continuously switches between HIGH and LOW, generating square pulses as a clock signal.
- Frequency and duty cycle depend on resistor and capacitor values:

$$f = \frac{1.44}{(R_A + 2R_B)C}$$

$$D = \frac{R_A + R_B}{R_A + 2R_B}$$

- Used as a continuous clock source for sequential circuits.

(b) Monostable Mode:

- Produces one pulse of fixed width when triggered by a push button.
- Pulse duration is given by:

$$T = 1.1 R C$$

- Used for manual or step-by-step clock control.

(c) Bistable Mode:

- Has two stable states (Set/Reset or 0/1).
- Output changes state only when triggered.
- Useful for enabling/disabling clock pulses or acting as a manual toggle switch.

3.2 Program Counter

The Program Counter (PC) is a 4-bit register in the SAP-1 computer that stores the address of the next instruction to be fetched from memory (RAM).

- The Program Counter outputs a 4-bit address
 - This address goes to the Memory Address Register (MAR) to fetch the corresponding instruction from RAM.
 - After each instruction fetch, the PC increments automatically by 1, pointing to the next instruction.
 - The 74LS245 bus transceiver is used to interface the PC with the system bus, ensuring controlled data flow.
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3.3 Integration of each module

At last, all previously built components—Clock, Program Counter (PC), Memory Address Register (MAR), RAM, A and B Registers, ALU, and the shared 8-bit data bus—were connected to form a complete working system.

- The system clock, generated using a 555-timer configured in astable mode, provides the timing pulses that synchronize the activity of every module.
 - The Program Counter (constructed from 74LS161 synchronous counters) increases its value with each clock pulse and places the address of the next instruction onto the bus. A 74LS245 transceiver ensures that the PC can drive the bus when needed.
 - The Memory Address Register receives the address from the bus and sends it directly to the RAM.
 - The RAM stores both instructions and data, and outputs the byte located at the provided address onto the shared data bus.
 - The A and B registers can read data from the bus when their load inputs are activated, allowing values from memory to be stored and used for calculations.
 - The Arithmetic Logic Unit (ALU) performs operations such as addition or subtraction using the values in the A and B registers. Depending on the control signals, the ALU's result can either be placed back on the bus or stored directly into Register A.
 - Since every module receives the same clock signal, data movement occurs in an orderly and controlled sequence.
 - An 8-bit bus (BUS 0–7) interconnects all modules. Bus direction and enable lines are controlled through 74LS245 buffers to prevent multiple outputs from driving the bus at the same time.
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4. Circuit Diagram

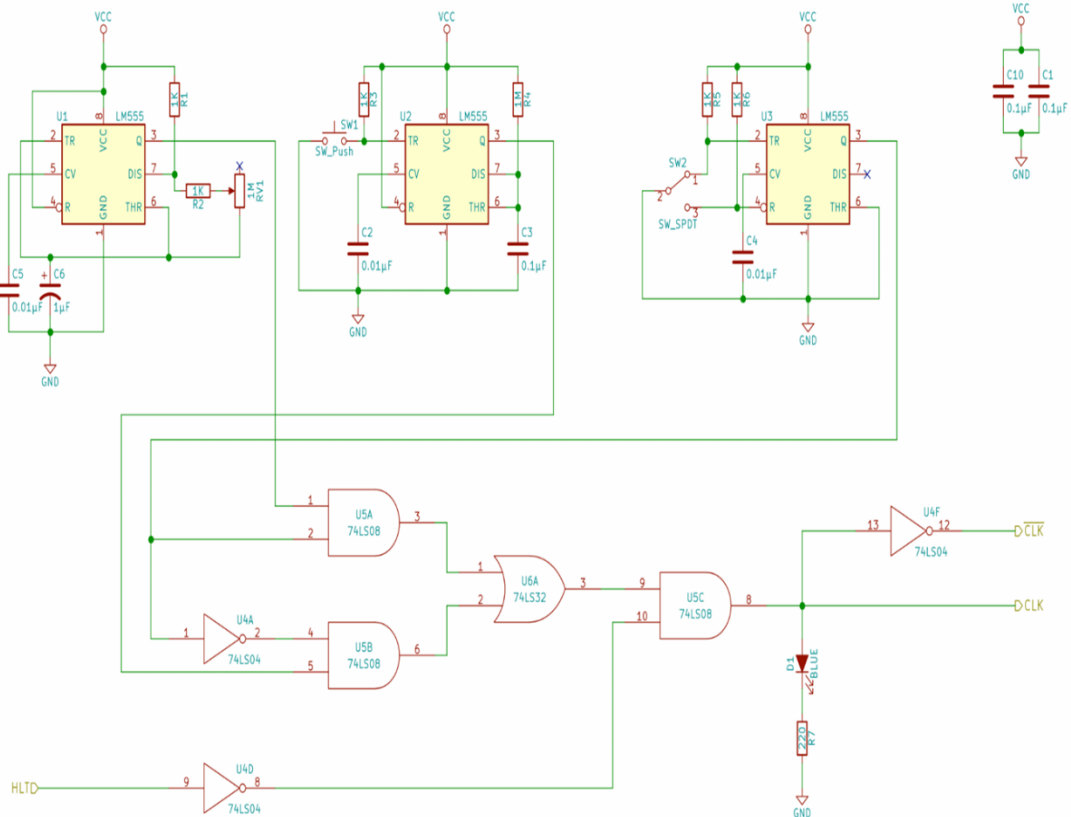


Fig. 1: Clock Schematic for three working modes: Astable, Monostable and Bistable

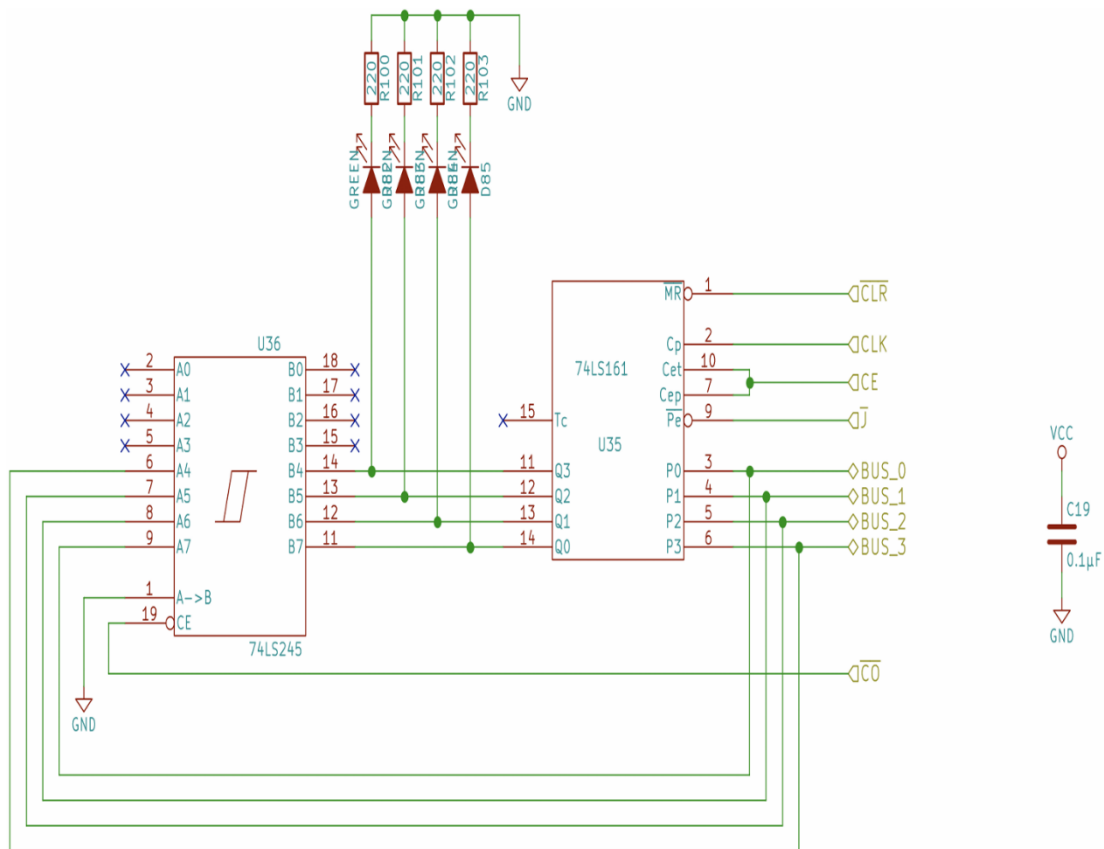


Fig. 2: Program Counter Schematic

5. Pin Diagram

5.1 LM555 Timer

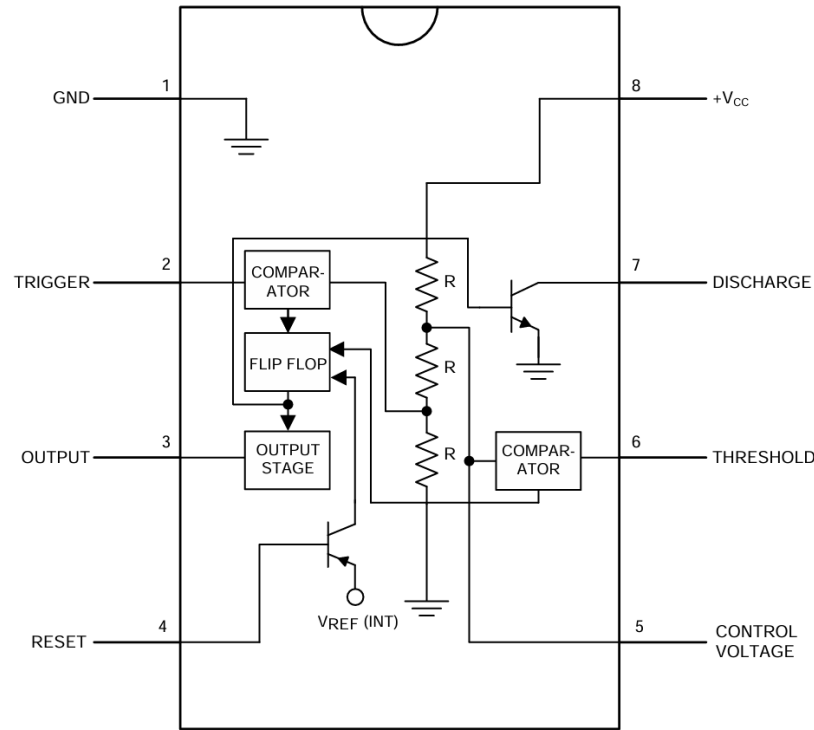


Fig. 3: LM555 Timer IC Pin Diagram

PIN	NAME	I/O	DESCRIPTION
5	Control Voltage	I	Controls threshold and trigger levels; determines pulse width of output waveform. External voltage here can modulate output waveform.
7	Discharge	I	Open collector output discharges capacitor between intervals; toggles output low when voltage reaches 2/3 of supply.
1	GND	O	Ground reference voltage.
3	Output	O	Output driven waveform.
4	Reset	I	Negative pulse resets timer; connect to VCC when unused to prevent false triggering.
6	Threshold	I	Compares input to 2/3 V _{CC} ; sets flip-flop state based on applied voltage.
2	Trigger	I	Causes flip-flop transition from set to reset; depends on external trigger pulse amplitude.
8	V ⁺	I	Supply voltage relative to GND.

5.2 4-Bit Synchronous Binary Counter (74LS161)

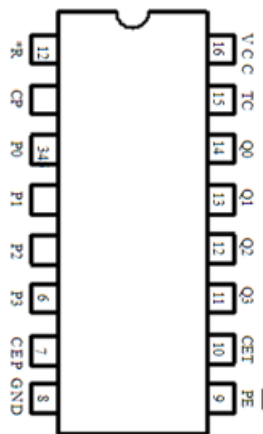


Fig. 4: 74LS161 Pin Diagram

PIN	NAME	I/O	DESCRIPTION
1	CEP (Count Enable Parallel)	I	Enables counting when both CEP and CET are HIGH.
2	P0	I	Parallel data input bit 0 (Least Significant Bit).
3	P1	I	Parallel data input bit 1.
4	P2	I	Parallel data input bit 2.
5	P3	I	Parallel data input bit 3 (Most Significant Bit).
6	CP (Clock Pulse)	I	Clock input; counter advances on rising edge when enabled.
7	CEP	I	Count Enable Parallel input; must be HIGH for counting.
8	GND	–	Ground (0V reference).
9	PE (Parallel Enable)	I	LOW enables parallel loading of P0–P3 into counter.
10	CET (Count Enable Trickle)	I	Enables ripple carry output (for cascading counters).
11	Q3	O	Output bit 3 (MSB).
12	Q2	O	Output bit 2.
13	Q1	O	Output bit 1.

14	Q0	O	Output bit 0 (LSB).
15	TC (Terminal Count)	O	HIGH when counter reaches maximum (15); used for cascading.
16	VCC	–	Supply voltage (+5V).

5.3 Octal Bus Transceiver (74LS245)

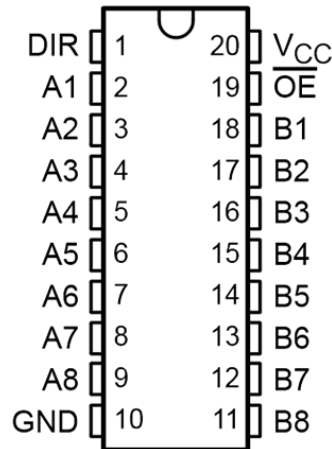


Fig. 4: 74LS245 Pin Diagram

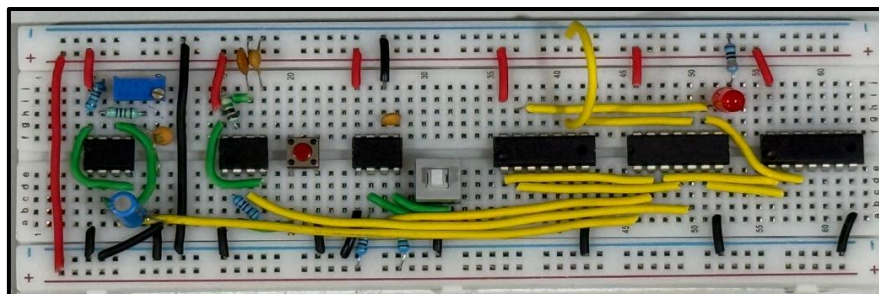
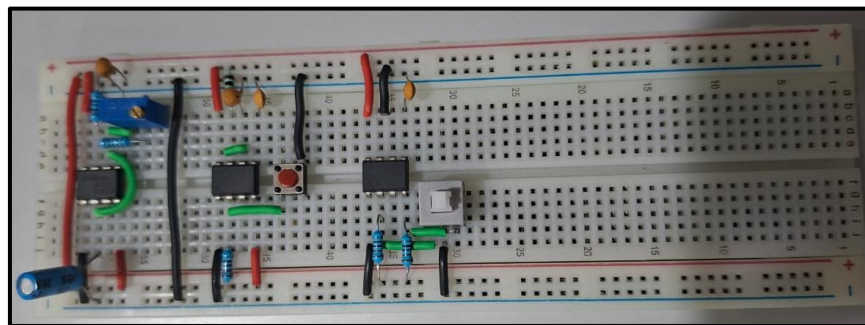
PIN	NAME	I/O	DESCRIPTION
1	DIR (Direction Control)	I	Controls the direction of data flow. HIGH → from A bus to B bus; LOW → from B bus to A bus.
2	A1	I/O	Data bus A, bit 1.
3	A2	I/O	Data bus A, bit 2.
4	A3	I/O	Data bus A, bit 3.
5	A4	I/O	Data bus A, bit 4.
6	A5	I/O	Data bus A, bit 5.
7	A6	I/O	Data bus A, bit 6.
8	A7	I/O	Data bus A, bit 7.
9	A8	I/O	Data bus A, bit 8.
10	GND	–	Ground (0V reference).
11	B8	I/O	Data bus B, bit 8.
12	B7	I/O	Data bus B, bit 7.
13	B6	I/O	Data bus B, bit 6.
14	B5	I/O	Data bus B, bit 5.
15	B4	I/O	Data bus B, bit 4.
16	B3	I/O	Data bus B, bit 3.

17	B2	I/O	Data bus B, bit 2.
18	B1	I/O	Data bus B, bit 1.
19	O $\bar{E}$ (Output Enable)	I	Active LOW enable; when HIGH, all outputs are in high-impedance (disabled) state.
20	VCC	–	Supply voltage (+5V).

6. Procedure

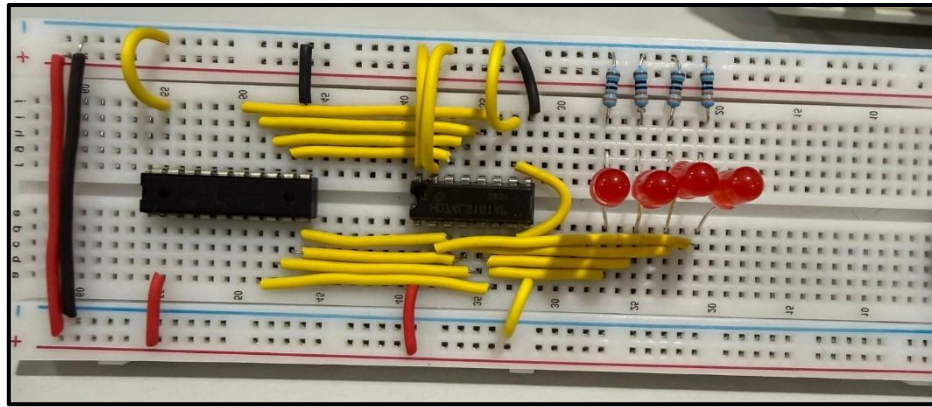
6.1 Assembling the Clock

1. Connected the 555 Timer IC on breadboard
 - Inserted the NE555 Timer IC onto the breadboard, ensuring proper orientation and connections.
2. Configured in Astable mode using resistors, capacitors and potentiometer.
 - Verified output waveform on DSO
 - Measured T_{ON} , T_{OFF} , T and duty cycle.
3. Configured Monostable mode using a push switch as trigger
 - Verified single pulse generation on DSO
4. Configured Bistable mode using a toggle switch
 - Verified switching between High and Low outputs.



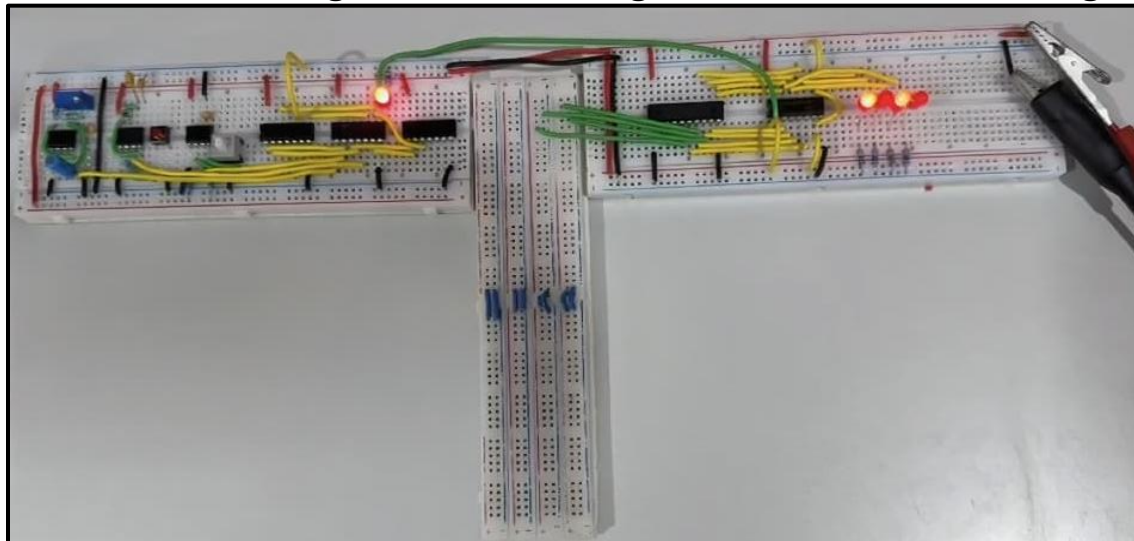
6.2 Assembling the Program Counter

- Implemented a 4-bit Program Counter (PC) using the 74LS161 synchronous counter for sequential address generation.
- Used the 74LS245 bus transceiver to interface the PC with the system bus, enabling controlled data transfer and preventing bus conflicts.



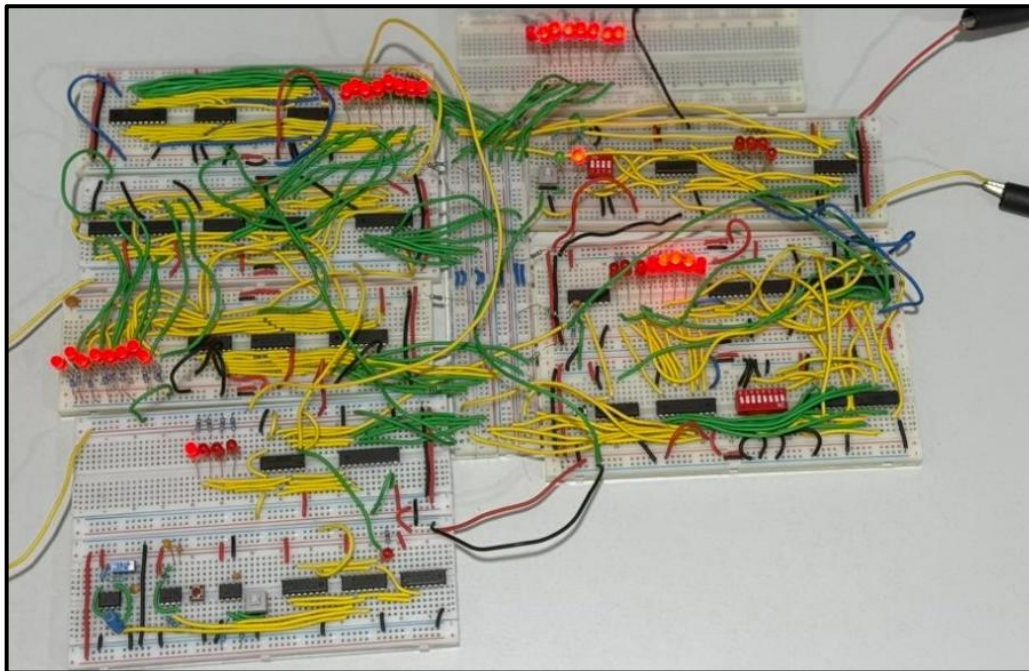
6.3 Integrating Program Counter and Clock

1. Interfacing the Program Counter with the Clock:
 - The Program Counter (PC) was connected to the output of the 555 Timer configured in Astable mode.
 - The PC's counting sequence was observed to advance in sync with the clock pulses.
2. Verification of Counting Operation:
 - Confirmed that the PC output incremented correctly from 0 to 15, advancing by one count per clock pulse.
3. Individual Module Testing:
 - The clock circuit and Program Counter were tested independently to verify proper functionality.
 - Debugging was carried out to resolve minor issues, ensuring accurate counting and correct bus interfacing.

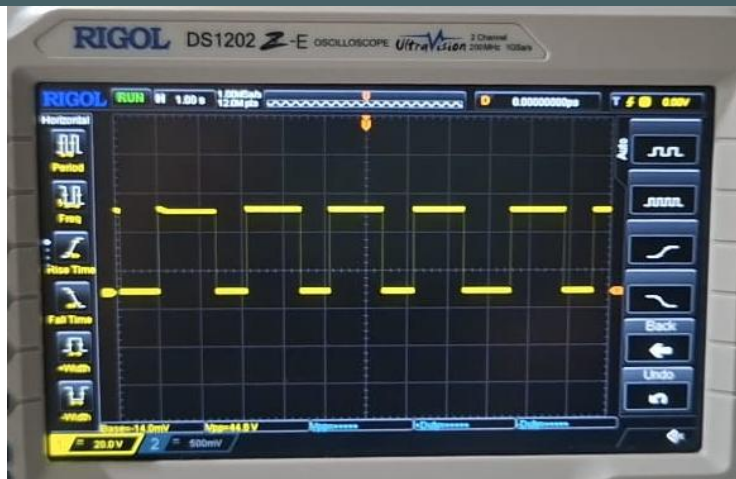


6.3 Integrating All Modules

1. The clock signal was distributed from the Clock circuit to the CLK pins of every module to ensure synchronized operation.
2. The Program Counter was connected to the Memory Address Register through a transceiver, allowing the PC to place address values onto the bus.
3. The output of the MAR was routed to the address inputs of the RAM module, while the data output of the RAM was tied to the system's 8-bit bus.
4. Both the A and B register's input were wired with the bus individually so they could receive data from the bus and provide their outputs as inputs to the ALU.
5. The result produced by the ALU was connected back to the bus, and also routed to the A register so results could be stored.
6. Verified that all data transfers and control signals synchronized with the common clock.
7. System behaviour—such as the Program Counter incrementing, MAR loading addresses, RAM fetching data, and ALU performing calculations—was monitored through LEDs and a digital oscilloscope.

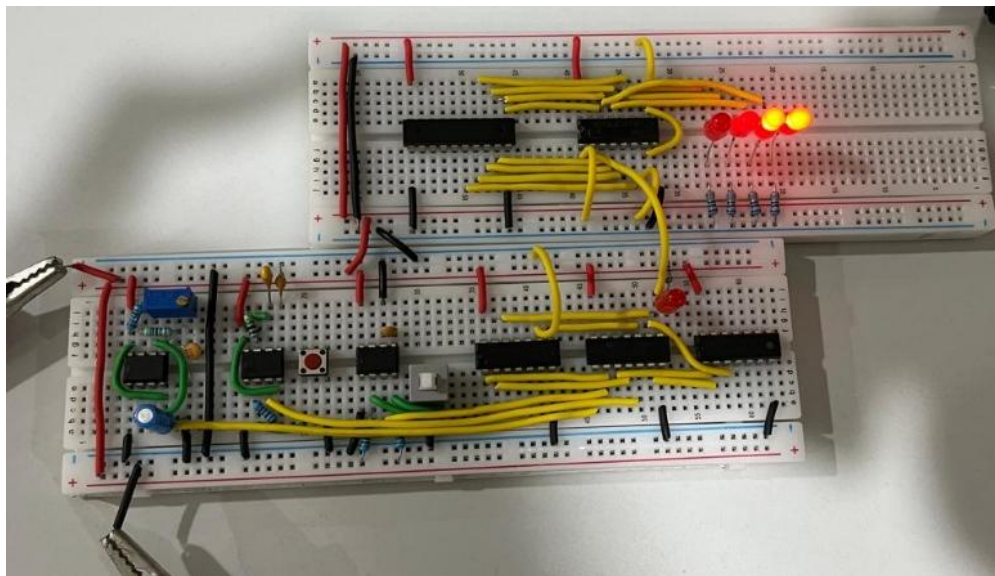


- ## 7.1 Clock Circuit



7.2 Program Counter (PC)

- The Program Counter incremented by 1 on each clock pulse, counting from 0 to 15 synchronously with the clock signal.
- The 74LS245 bus transceiver correctly interfaced the PC output with the bus, ensuring no data conflicts with other components.
- The Program Counter operated correctly with the clock in Astable mode, demonstrating proper synchronization between the clock and the counting operation.



7.3 Integrated 8-Bit Computer

- Every module operated from a common clock signal, ensuring they all worked in step.
 - The Program Counter successfully transferred its address to the Memory Address Register, which then provided the address to the RAM.
 - The RAM output appeared correctly on the 8-bit bus, and both the A and B registers were able to load (latch) the data from the bus when enabled.
 - The ALU executed arithmetic operations (addition and subtraction) using the register values and placed the computed result back onto the bus and into Register A as required.
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8. Conclusion

- A versatile clock system was built using a 555 timer IC, capable of operating in Astable, Monostable, and Bistable modes, allowing different timing behaviours depending on the requirement.
 - This clock circuit successfully generated stable timing pulses and served as the primary clock source for the entire system.
 - The Program Counter (PC) was implemented using 74LS161 synchronous counters, along with 74LS245 transceiver to ensure controlled data flow onto the system bus.
 - The PC reliably generated sequential memory addresses.
 - Both the clock module and the Program Counter were tested on a breadboard.
 - After individual validation, all major modules—Clock, PC, Memory Address Register (MAR), RAM, A and B Registers, ALU, and the shared 8-bit bus—were interconnected.
 - Every module received the same clock pulse, ensuring synchronized operation throughout the system.
 - Data moved correctly along the bus: the PC provided an address to the MAR, which then accessed the RAM, and the registers and ALU correctly performed data transfers and arithmetic functions.
 - The fully assembled setup demonstrated a complete and functioning data path for the 8-bit computer.
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